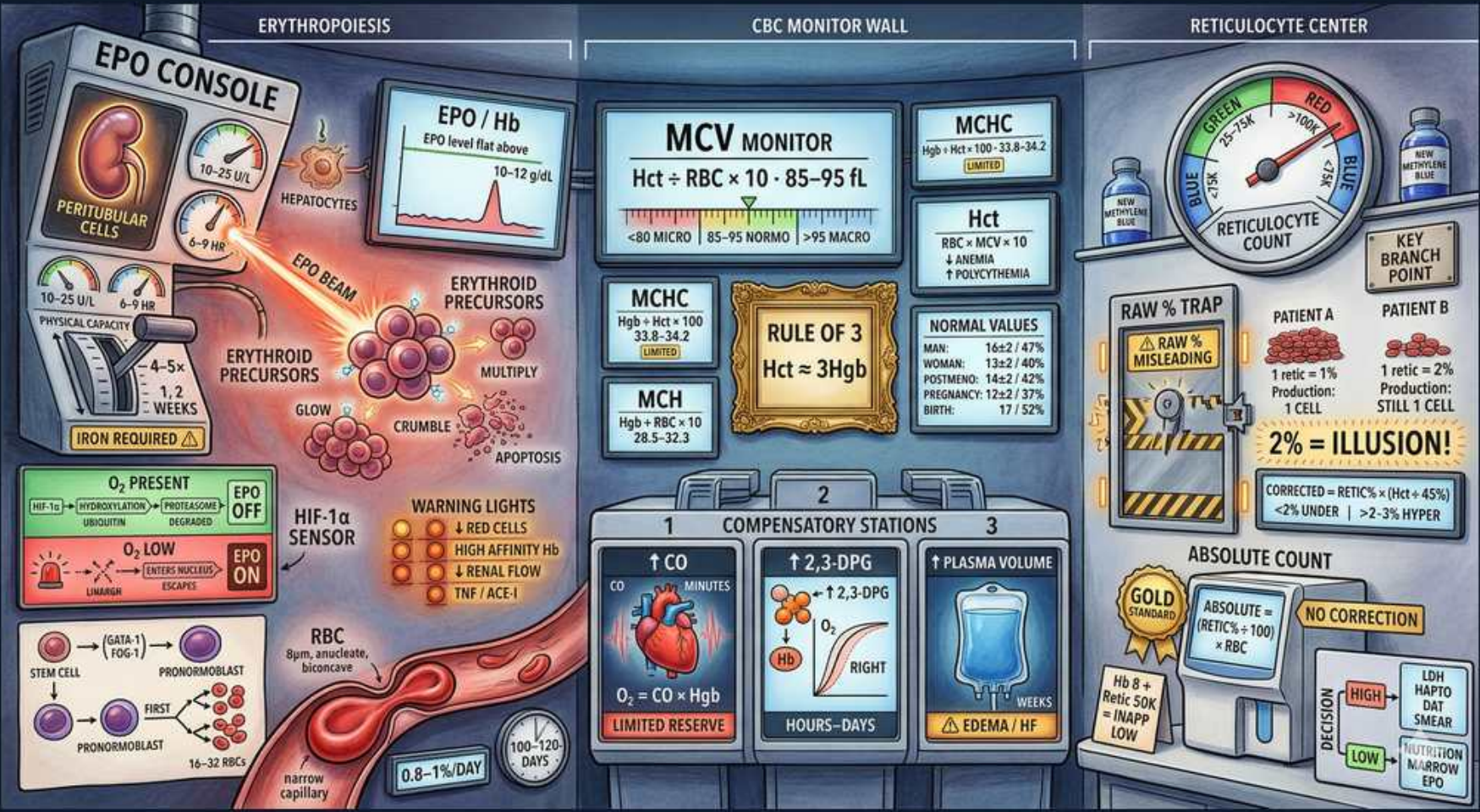


Anemia Workup — Reticulocytes & RBC Indices



1. EPO from kidney

WHAT YOU SEE

The 'EPO CONSOLE' built into a kidney, with peritubular cells firing an EPO beam — the kidney is drawn as the control panel because ~90% of erythropoietin is made there.

WHAT IT ENCODES

Erythropoietin (EPO) is made by renal peritubular interstitial (fibroblast-like) cells in the cortex in response to tissue hypoxia; it binds the EPO receptor on erythroid progenitors to drive proliferation and survival. EPO is the rate-limiting signal for erythropoiesis. About 90% of the body's EPO is renal (the remainder hepatic), which is why advanced CKD blunts the EPO response.

BOARD TRAP

WRONG: 'Anemia of CKD is treated by giving iron alone because the marrow just lacks substrate.' The core defect in CKD anemia is inadequate renal EPO production (hypoproliferative, low retic). Discriminator: CKD anemia needs an erythropoiesis-stimulating agent (epoetin/darbepoetin) once iron stores are repleted (target ferritin/TSAT first); iron alone won't fix EPO deficiency.

2. HIF-1α O2 sensor

WHAT YOU SEE

The 'HIF-1α SENSOR' panel with warning lights: when O2 drops the sensor flips EPO ON — HIF is the oxygen detector that switches on erythropoietin in hypoxia.

WHAT IT ENCODES

The oxygen sensor is HIF (hypoxia-inducible factor). In normoxia, prolyl hydroxylases (PHD) hydroxylate HIF-1α/2α, allowing VHL-mediated ubiquitination and degradation. In hypoxia, PHDs are inactive, HIF stabilizes and transcribes EPO. This is the target of HIF-PH inhibitors (e.g., roxadustat) for CKD anemia, and VHL loss explains EPO-secreting hemangioblastoma/RCC.

BOARD TRAP

WRONG: 'Chuvash polycythemia and VHL-related erythrocytosis are caused by primary marrow JAK2 mutations.' These are EPO-axis (HIF/VHL) disorders causing inappropriately high EPO, not autonomous JAK2-driven marrow disease. Discriminator: HIF/VHL pathway lesions give HIGH EPO; PV gives LOW EPO with JAK2 mutation.

3. MCV 80-95

WHAT YOU SEE

MCV monitor: Hct ÷ RBC ×10, 80-95 fL

WHAT IT ENCODES

Mean corpuscular volume MCV = (Hct ÷ RBC count) × 10, normal ~80-100 fL (board shorthand 80-95). It is the primary morphologic axis: microcytic <80 (iron deficiency, thalassemia, anemia of chronic disease, sideroblastic), normocytic 80-100 (acute blood loss, hemolysis, early/mixed), macrocytic >100 (B12/folate, alcohol/liver, hypothyroid, MDS, drugs).

BOARD TRAP

WRONG: 'A normal MCV rules out a clinically important nutritional anemia.' A combined iron deficiency (micro) + B12/folate deficiency (macro) can produce a falsely NORMAL average MCV with a high RDW. Discriminator: a normal MCV with a wide RDW or a dimorphic smear should prompt looking for a mixed deficiency.

4. MCHC

WHAT YOU SEE

A CBC monitor screen reading 'MCHC = Hgb ÷ Hct' (32-36) — the concentration index that runs HIGH in spherocytosis and LOW in iron deficiency.

WHAT IT ENCODES

Mean corpuscular hemoglobin concentration MCHC = Hgb ÷ Hct (g/dL of Hb per dL of cells), normal ~32-36. HIGH MCHC (hyperchromic) is classic for hereditary spherocytosis (and a clue/artifact in cold agglutinins or autoagglutination). LOW MCHC (hypochromia) is seen in iron deficiency and thalassemia. It is one of the few indices that can be truly elevated.

BOARD TRAP

WRONG: 'A high MCHC always means hereditary spherocytosis.' A spuriously high MCHC also occurs with cold agglutinin disease, lipemia, or marked hyperglycemia (RBC clumping/measurement artifact). Discriminator: confirm spherocytes on smear and a positive osmotic fragility/EMA-binding test before attributing a high MCHC to HS.

5. Rule of 3

WHAT YOU SEE

RULE OF 3: Hct ≈ 3×Hgb

WHAT IT ENCODES

The 'rule of three': Hct (%) ≈ 3 × Hgb (g/dL), and Hgb ≈ 3 × RBC count (millions). It is a quick internal consistency check on the CBC. A mismatch suggests a lab artifact or abnormal RBCs (e.g., spuriously high MCHC from lipemia/cold agglutinins, hyperglycemia-induced cell swelling, or paraprotein interference).

BOARD TRAP

WRONG: 'If the rule of three fails, just trust the Hgb and ignore it.' A broken rule of three usually flags an analytic interference (lipemia, cold agglutinins, severe leukocytosis, hyperglycemia) that distorts measured indices. Discriminator: investigate/repeat the sample (warm it for cold agglutinins, clear lipemia) rather than acting on the discordant value.

6. Reticulocyte count

WHAT YOU SEE

The big central 'RETICULOCYTE COUNT' gauge flagged 'KEY BRANCH POINT' — the dial that splits anemia into marrow-responding vs marrow-failing.

WHAT IT ENCODES

The reticulocyte response is the single most useful early test in anemia because it tells you whether the marrow is responding. It splits all anemia into hyperproliferative (appropriate marrow response — hemolysis or acute blood loss) versus hypoproliferative (inadequate response — nutritional, marrow, or EPO problem), guiding the entire downstream workup before any smear or iron studies.

BOARD TRAP

WRONG: 'In any anemia, start with iron studies and B12 before checking reticulocytes.' The reticulocyte count is the correct first branch point — it separates production failure from destruction/loss and prevents unnecessary testing. Discriminator: a high corrected retic/RPI redirects you to hemolysis/blood-loss workup, not iron/B12.

7. Raw % misleading

WHAT YOU SEE

The 'RAW % MISLEADING' warning panel showing '2% = ILLUSION' — a cartoon trap warning that the raw retic percent is inflated by the shrunken RBC denominator in anemia.

WHAT IT ENCODES

The raw reticulocyte percentage is a fraction of total RBCs, so when the RBC count falls (anemia), the SAME number of retics looks like a higher percentage — an artifactual elevation. You must correct for the degree of anemia. This is why a '2% retic' in a severely anemic patient may actually represent an inadequate marrow response.

BOARD TRAP

WRONG: 'Retic of 4% means the marrow is responding well, so this anemia is hemolytic.' The raw percent is inflated by the low RBC denominator; once corrected for Hct (and shift), it may be inappropriately low. Discriminator: use the corrected retic count or reticulocyte production index, not the raw percentage.

8. Corrected retic

WHAT YOU SEE

Corrected = retic% × Hct/45

WHAT IT ENCODES

Corrected reticulocyte count = retic% × (patient Hct ÷ 45). When EPO is high, immature retics ('shift cells') are released early and live longer in blood, so the Reticulocyte Production Index (RPI) further divides by a maturation factor (~1 at Hct 45 up to ~2.5 at Hct ~15). RPI >2-3 = adequate marrow response (hemolysis/blood loss); RPI <2 = hypoproliferative.

BOARD TRAP

WRONG: 'Once you correct the retic for Hct you never need the maturation correction.' In severe anemia early-release shift reticulocytes overstate effective production; the RPI maturation factor corrects this. Discriminator: in marked anemia, compute the RPI (divide by the shift factor) — RPI <2 still means an inadequate marrow response.

9. Absolute count gold

WHAT YOU SEE

The 'ABSOLUTE COUNT' box stamped GOLD STANDARD: retic% × RBC — the gold tag marks it as the most reliable measure of true marrow output (no denominator distortion).

WHAT IT ENCODES

Absolute reticulocyte count = retic% × RBC count (normal ~25,000-75,000/μL, or expressed as ~0.025-0.075 × 10⁶). Because it uses the actual RBC number rather than a percentage, it is the most reliable single measure of marrow output and is not distorted by the anemia denominator. A high absolute retic confirms an effective erythropoietic response.

BOARD TRAP

WRONG: 'A normal absolute reticulocyte count in a hemolyzing patient is reassuring.' In active hemolysis the marrow should ramp up to 4-6× baseline; a merely 'normal' absolute retic is inappropriately low and suggests a superimposed problem (e.g., parvovirus B19 aplastic crisis, coexisting iron/folate deficiency). Discriminator: judge the retic against the expected response, not the reference range.

10. Low retic = marrow

WHAT YOU SEE

The DECISION arrow reading 'LOW retic → marrow / iron / EPO problem' — a low dial points away from destruction toward a production failure.

WHAT IT ENCODES

A low/inadequate reticulocyte response (RPI <2, low absolute count) = hypoproliferative anemia: a production problem from missing substrate (iron, B12, folate), missing signal (low EPO in CKD), or marrow failure/infiltration (aplastic anemia, MDS, leukemia, fibrosis, parvovirus B19). Next steps are iron studies, B12/folate, renal function, and—if unexplained—bone marrow biopsy.

BOARD TRAP

WRONG: 'A low reticulocyte count with macrocytosis is most likely hemolysis.' Low retics indicate failed production, not destruction; macrocytosis with low retics points to B12/folate deficiency or MDS. Discriminator: hemolysis raises retics (RPI >2) with high LDH/indirect bili and low haptoglobin — the opposite of this picture.

11. ↑CO compensation

WHAT YOU SEE

Compensatory Station 1 showing a heart pumping harder ('↑CO') with 'LIMITED RESERVE' — the first, fastest defense against anemia is raising cardiac output, but the reserve is finite.

WHAT IT ENCODES

The first physiologic compensation for anemia is increased cardiac output (tachycardia, increased stroke volume) to maintain O₂ delivery, occurring within seconds-minutes. Reserve is limited—at very low Hgb (≈<7) or in patients with coronary disease this high-output state can precipitate demand ischemia and high-output heart failure.

BOARD TRAP

WRONG: 'A young healthy patient can compensate for acute massive blood loss indefinitely with increased cardiac output.' The cardiac-output reserve is finite; ongoing loss leads to decompensation/shock, and in chronic anemia sustained high output can cause failure. Discriminator: tolerance depends on rate of loss and cardiac reserve, not Hgb number alone.

12. ↑2,3-DPG

WHAT YOU SEE

Compensatory Station 2 showing the O₂-dissociation curve shifted RIGHT by rising 2,3-DPG over hours-days — more 2,3-DPG unloads O₂ to hypoxic tissue.

WHAT IT ENCODES

Over hours-to-days, erythrocyte 2,3-bisphosphoglycerate (2,3-DPG/2,3-BPG) rises, binding deoxyhemoglobin and shifting the oxygen-hemoglobin dissociation curve to the RIGHT. This lowers Hgb's O₂ affinity (raises P₅₀) so more O₂ is unloaded to hypoxic tissues at the same PO₂ — a key adaptive compensation in chronic anemia.

BOARD TRAP

WRONG: 'Increased 2,3-DPG shifts the oxygen curve LEFT to hold onto oxygen.' 2,3-DPG, along with acidosis, fever, and high CO₂, shifts the curve RIGHT (decreased affinity, enhanced tissue unloading). Discriminator: LEFT shifts come from low 2,3-DPG, alkalosis, hypothermia, carboxyhemoglobin, fetal Hb — the opposite stimuli.

13. ↑plasma volume

WHAT YOU SEE

Compensatory Station 3 showing an expanding plasma/IV bag over weeks with an EDEMA/HF caution — plasma volume swells to refill the circulation, diluting Hct and risking overload.

WHAT IT ENCODES

Over weeks, the body expands plasma volume to restore circulating volume after red cell loss. This dilutes the remaining RBCs (lowering measured Hgb/Hct without further true red cell loss) and, in patients with cardiac/renal impairment, can precipitate volume overload, edema, and heart failure. It also explains why Hct underestimates red cell mass in chronic states.

BOARD TRAP

WRONG: 'A falling Hct over weeks after a treated bleed always means rebleeding.' Plasma volume expansion dilutes the Hct (hemodilution) as part of normal compensation, lowering Hct without new blood loss. Discriminator: stable vitals, rising reticulocytes, and a hemodynamically quiet patient favor dilution over active rebleeding.

14. EPO → marrow

WHAT YOU SEE

Erythroid precursors multiplying and maturing under the EPO beam in the marrow panel — EPO drives the colonies to grow, but only if iron/B12/folate substrate is present.

WHAT IT ENCODES

EPO drives erythroid colony-forming units to proliferate and mature (proerythroblast → reticulocyte), but the marrow can only respond if it has substrate. Effective erythropoiesis therefore requires adequate iron, B12, and folate; the normal marrow can ramp output up to ~6-8× baseline when fully stimulated and supplied. Lack of substrate produces ineffective erythropoiesis despite high EPO.

BOARD TRAP

WRONG: 'Giving exogenous EPO will correct any hypoproliferative anemia.' EPO is useless if the marrow lacks iron (functional iron deficiency), B12/folate, or is infiltrated/aplastic. Discriminator: replete iron stores (and B12/folate) and exclude marrow failure before or alongside ESA therapy — substrate plus signal are both required.

15. MCH

WHAT YOU SEE

A sub-readout on the CBC monitor wall labelled 'MCH' beneath the MCHC screen — the per-cell hemoglobin content gauge ($\text{Hb} \div \text{RBC}$), normal ~27-33 pg.

WHAT IT ENCODES

Mean corpuscular hemoglobin $\text{MCH} = \text{Hgb} \div \text{RBC count} (\times 10)$, the average mass of hemoglobin PER red cell, normal ~27-33 pg. It largely tracks MCV (small cells carry less Hb), so it falls in iron deficiency and thalassemia (hypochromic, microcytic) and rises in macrocytic anemias. Unlike MCHC (a concentration), MCH is an absolute amount per cell and is less useful for detecting spherocytosis.

BOARD TRAP

WRONG answer: treating a low MCH as a distinct diagnosis or confusing it with MCHC. Discriminator: MCH is hemoglobin amount PER cell (pg) and parallels MCV, whereas MCHC is hemoglobin CONCENTRATION (g/dL) and is the index that runs truly HIGH in spherocytosis — use MCV/RDW and iron studies, not MCH alone, to separate IDA from thalassemia.